

MS MARCO Generative QA — Progress Report

Generated 2026-05-14 15:02. For per-week depth see reports/generated/weekNN_.pdf.*

Project goal

Build an end-to-end open-domain QA system on MS MARCO that goes **query** → **retrieve passages** → **generate an answer**, and measure each stage in isolation so it's clear where errors come from. The aim is not SOTA; the aim is a clean, reproducible pipeline whose retrieval, reranking, and generation stages can each be ablated.

Methods (by week)

Week	Stage	Data	Method
W1	EDA	MS MARCO QA v2.1 validation, 5k-row sample	Length / type / answer-type distributions; sizing decisions for W2–W5
W2	Sparse retrieval (BM25)	MS MARCO Passage (8.8M docs), dev/small (6 980 q)	bm25s backend, $k_1=1.5$, $b=0.75$, top-1000 per query
W3	RAG generation	BM25 top-3 from $W2 \cap QA$ references, 200 q	T5-small (no fine-tuning), prompt = question: <q> context: <p1 p2 p3>
W4	Dense retrieval	Qrels-anchored 50k sample of Passage, dev/small (6 980 q)	all-MiniLM-L6-v2 + FAISS IndexFlatIP; same sample re-indexed with BM25 for apples-to-apples comparison
W5	Cross-encoder reranking	W4 dense top-100, full dev/small (6 980 q)	cross-encoder/ms-marco-MiniLM-L-6-v2 rerank, depth $K=100$

All runs share a single config (configs/baseline.yaml); the four entrypoints in experiments/run_*.py produce the metrics each report cites.

Key results

W1 (EDA) — Queries are short (5–10 words median); passages $\sim 10\times$ longer (median ~ 50 words). DESCRIPTION and NUMERIC dominate the query-type buckets; a non-trivial fraction of queries has no answer and must be filtered for any SFT supervision. Most queries have 0 or 1 marked-relevant passage (sparse, binary qrels).

W2 (BM25 baseline, full 8.8M corpus, 6 980 dev/small queries)

Metric	Value
MRR@10	0.1703
Recall@100	0.6212
Recall@1000	0.8154

Published Anserini/Lucene reference is ~ 0.184 MRR@10; the ~ 0.014 gap is attributable to tokenizer differences between Lucene and bm25s. No k1/b tuning. Search cost: 586 ms/query.

W3 (RAG, T5-small + BM25 top-3, 200 queries)

Metric	Value
ROUGE-L	0.1619
BLEU	0.0573
EM	0.0050
Token F1	0.1756

Honest end-to-end check, not a leaderboard number. T5-small is pretrained-only (no SFT) and produces verbose, extractive-style outputs that are penalised by surface-form metrics. Retrieval errors propagate directly into generation — a reranking step (W5) is expected to lift these numbers when wired in.

W4 (Dense vs BM25 on the same 50k qrels-anchored sample, 6 980 queries)

Metric	BM25 (sample)	Dense (sample)	Δ
MRR@10	0.6948	0.8830	+0.1882
nDCG@10	0.7270	0.9041	+0.1771
Recall@100	0.9338	0.9946	+0.0608
Recall@1000	0.9686	0.9991	+0.0305

Dense wins on every metric. Caveat: every dev/small relevant doc is unconditionally included in the 50k pool, so absolute numbers are upper-bounded; the *delta* between dense and BM25 on the same pool is what's meaningful. Recall@100 = 0.9946 means dense has effectively hit the ceiling on this sampled setting $\rightarrow$ from W5 onward, recall is no longer the bottleneck.

W5 (Cross-encoder rerank over W4 dense top-100, full 6 980-query dev/small)

Metric	Dense (top-100)	Dense + CE rerank	Δ
MRR@10	0.8830	0.9304	+0.0474
nDCG@10	0.9041	0.9434	+0.0393
Recall@100	0.9946	0.9946	+0.0000

Recall@100 is unchanged by construction (the reranker only re-orders the top-100). The interesting columns are MRR@10 and nDCG@10: even with recall saturated, the cross-encoder still buys roughly +0.04 by tightening *local ordering* — pushing the relevant passage from rank 2–3 to rank 1. Cost: 4 h 37 min wall-clock for 538 k pairs (~32 pairs/s), peak RSS 3.3 GiB; the CE is the slowest stage by ~2 orders of magnitude. The earlier 1 000-query subsample of dev/small produced consistent deltas (MRR Δ +0.0435, nDCG Δ +0.0398), confirming the gain is not an artefact of the subsample.

Limitations

- **Sampled corpus from W4 onward.** Dense + rerank numbers are on a 50k qrels-anchored sample where every relevant doc is present by construction. Comparable *within* W4/W5; the W4/W5 absolute MRR is *not* directly comparable to the W2 full-corpus number.
- **No fine-tuning anywhere.** Generic encoders/generator used as-is (all-MiniLM-L6-v2, ms-marco-MiniLM-L-6-v2, t5-small). A MS-MARCO-tuned encoder (msmarco-MiniLM-L6-cos-v5) or SFT generator would close most of the gap to leaderboard baselines.
- **Dev/small only, single seed.** No held-out test split; distractor sample is deterministic but not averaged across seeds.
- **W3 was evaluated on 200 queries.** A subsample of dev/small for compute reasons; metric variance not quantified. W5 now covers full dev/small (6 980 queries).
- **Surface-form generation metrics only.** ROUGE/BLEU/EM penalise valid paraphrases — a semantic metric (BERTScore) would give a fairer signal.

Next steps

- Wire W5's reranked top-K into the W3 RAG generator and measure whether better ordering improves answer quality (not just retrieval metrics).
- Rerank BM25 top-100 (W2 run) and compare the delta to the dense delta — does the reranker recover more from a weaker first stage?
- Hybrid first-stage (RRF over BM25 + dense) → rerank.
- Scale the dense sample from 50k to 200k–500k passages to track how the BM25[?]dense gap shifts as relevant-doc density drops.
- Add a held-out test split and BERTScore for generation, so improvements aren't overfit to dev/small or to surface-form metrics.

How this maps to the repo

- Conclusions per week: reports/generated/weekNN_*.pdf
- Process logs per week: notebooks/weekNN_*.ipynb

- Reproducible pipeline: experiments/run_*.py + src/ + configs/
baseline.yaml
- Reproduce locally: make targets (see Makefile); outputs land in outputs/ (not committed).